Niklas Erdmann



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linkedin, google scholar, homepage, orcid, github

About Me: I am a machine learning researcher completing a PhD at the University of Oslo (expected 04.2027) on probabilistic forecasting for dynamic energy systems. I develop monotonic deep lattice networks that forecast full predictive distributions over multiple horizons, with validity guaranteed by the architecture. I want to extend shape-constrained neural networks towards physics-informed and trustworthy AI, and I enjoy teaching and supervising students alongside research.

Post-Graduate Education:

PhD, Dep. of Technology Systems, University of Oslo **03.2023 - 04.2027**
Title: *Artificial Intelligence-based Prognosis of Dynamic Energy Systems*

I work on probabilistic forecasting and uncertainty quantification for energy time series, from solar irradiance forecasting and nowcasting to battery state-of-health estimation. The project is a collaboration with the Institute for Energy Technology (IFE) and the Norwegian Defence Research Establishment (FFI).

Publications

Erdmann, N., Bentsen, L., Stenbro, R., Riise, H. N., Warakagoda, N. D., & Engelstad, P. E. (2026). Multi-Horizon Time Series Forecasting of Non-Parametric CDFs with Deep Lattice Networks. In *Proceedings of the 40th Annual AAAI Conference on Artificial Intelligence.* (arXiv, Code and Data)

Eriksen, E. W., Nygård, M. M., Erdmann, N., & Riise, H. N. (2026). Deep Learning Multihorizon Irradiance Nowcasting: A Comparative Evaluation of Three Methods for Leveraging Sky Images. *IEEE Journal of Photovoltaics*, *16*(4), 594–604. (DOI)

Erdmann, N., Bentsen, L. Ø., Stenbro, R., Riise, H. N., Warakagoda, N., & Engelstad, P. (2024). Deep and Probabilistic Solar Irradiance Forecast at the Arctic Circle. In *Proceedings of the 2024 IEEE 52nd Photovoltaic Specialist Conference (PVSC)* (pp. 1599–1606). (IEEE, arXiv; Best Poster Award)

Fordham, D., & Erdmann, N. (2026). Estimating Lithium-Ion Battery State-of-Health with Monotonic Lattice Networks. *SSRN preprint*. (DOI)

Zieglmeier, S., Erdmann, N., & Warakagoda, N. D. (2025). Reinforcement Learning for Pollution Detection in a Randomized, Sparse and Nonstationary Environment with an Autonomous Underwater Vehicle. *arXiv preprint arXiv:2510.26347*. (arXiv)

Education:

MSc. Artificial Intelligence, University of Groningen **09.2019 - 07.2022**
English-taught, Grade: 8.4/10, ECTS: 121
Master Thesis: *"Dynamic Modeling of a P(VDF-TrFE-CTFE)-Composite Soft Actuator"* (RUG): dynamic modelling of an electro-active actuator from noisy experimental time series with echo state networks.

Completed courses across the AI spectrum: Data Science, Machine/Deep Learning, Computer Vision, Pattern Recognition, Multi-Agent Systems, Reinforcement Learning, and Neuroscience.

Publication

D'Anniballe, R., Erdmann, N., Selleri, G., & Carloni, R. (2022). Dynamic Modeling of P (VDF-TrFE-CTFE)-based Soft Actuators via Echo State Networks. In *Proceedings of the 2022 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)* (pp. 118–124). (IEEE)

BSc. Psychology, University of Groningen **09.2015 - 01.2019**
English-taught, Grade: 7.6/10, ECTS: 200

Excellence track of the Honours Programme, with a focus on Statistics, Research Methods, and Cognitive Psychology, and a minor in Artificial Intelligence. Followed by a Pre-Master in Artificial Intelligence (09.2018 - 08.2019, Grade 8.1/10).

Teaching Experience:

PhD Teaching Component, University of Oslo **09.2024 - 04.2027**
Lectures, assignment design and exams for TEK5040 Deep Learning for Autonomous Systems. Independent supervision of 5 Master's thesis projects (3 completed, 2 ongoing).

Teaching Assistant, University of Groningen **09.2019 - 07.2021**
Led tutorial sessions, graded assignments and exams, and coordinated a group of other TAs.

Teaching Qualifications:

Basic Competence in University Pedagogy, UiO **expected 04.2027**
Foundation module including peer mentoring (120 h) and the Research Supervision elective (30 h); the formal teaching qualification for academic staff at Nordic universities.

Teaching and Learning in STEM, UiO **2025**
30 h course on constructive alignment, backward design and the scholarship of teaching and learning.

Other Activities:

Talks and Posters

Poster (on site) and online talk, 40th AAAI Conference on Artificial Intelligence, Singapore (2026).
Invited talk, "Forecasting Uncertainty with ML", FME Solar Energy (2025).

Awards

Best Poster Award, IEEE 52nd Photovoltaic Specialist Conference (PVSC) (2024).
Best Paper Award, PhD educational component, University of Oslo, for the AUV pollution-detection paper.
Best Performer: Leukemia Challenge, biomarker classification, MSc AI, University of Groningen.

Academic Service

PhD/PostDoc Representative, Department Board, Dep. of Technology Systems (2025); vice-representative, Faculty Board, Faculty of Mathematics and Natural Sciences (2024), University of Oslo.

Relevant Skills and Experiences:

Languages
German (Native), English (Professional; fully English-taught education since 2015), Dutch & Norwegian (Elementary).

Programming
Python (primary) with PyTorch, TensorFlow, SciPy, Optuna and OpenCV.
Also: C, R, Matlab; Git, LaTeX, Slurm (HPC).
Agentic coding for research software (Claude Code).

References: Available upon request.